Gene Name Shortcuts Inflate AI and Language Model Benchmarks of Pathway Knowledge

Muhammad Azam[1] and Michael Olaolu Arowolo[2]

[1]Center for Precision Health, McWilliams School of Biomedical Informatics,
The University of Texas Health Science Center at Houston, Houston, TX, USA
[2]M.S. Health Informatics Program, Department of Public Health Sciences,
Xavier University of Louisiana, New Orleans, LA 70125, USA

Muhammad.Azam@uth.tmc.edu, marowolo@xula.edu

***Abstract*—Curated pathway databases such as KEGG are used as ground truth to judge whether artificial intelligence (AI) systems, including large language models (LLMs), can recover the type of a gene–gene regulatory relation. We show that this design rewards shortcuts. Auditing a public KEGG-derived benchmark of 3,207 relations from 23 human pathways, we find that 19.6% of gene pairs carry more than one label, 41.0% of random-split test rows share their gene pair with training, and gene symbols encode function through family stems. A classifier reading only the two gene symbols reaches a macro-F1 of 0.731 under the random-row protocol, within the range reported for LLMs. We reformulate the task as pair-level multi-label prediction and evaluate seven machine-learning predictors and four pre-trained biomedical language models (BioBERT, SciBERT and SPECTER; frozen and fine-tuned) under four splits of increasing strictness. Macro AUROC falls from 0.99 when only pairs are held out to 0.81 for unseen gene families and about 0.70 for unseen pathways. The collapse is selective: phosphorylation remains predictable in unseen pathways (AUROC 0.87) because kinase stems are learnable, whereas the activation/inhibition sign falls to 0.56–0.65, although the sign is consistent across pathways for 83% of shared pairs. Frozen language models only match the symbol-only baseline on unseen pathways (AUROC 0.70–0.71 against 0.68), fine-tuned BioBERT gains modestly (0.73) while the regulatory sign stays below 0.68, and scrambling gene names removes 0.20–0.23 of the language models' unseen-gene AUROC. High random-split scores measure familiarity with gene names rather than pathway knowledge. We release all splits and code.**

***Index Terms*—Artificial intelligence, benchmark design, biomedical knowledge mining, data leakage, gene–gene interaction, graph neural networks, KEGG, language models, BioBERT, large language models, shortcut learning.**

I. Introduction

SIGNALING pathways encode which gene products activate, inhibit or phosphorylate one another, and they are a basic substrate for mechanistic reasoning in precision oncology, pharmacology and systems biology. The Kyoto Encyclopedia of Genes and Genomes (KEGG) [1], [2] is among the most widely used manually curated sources of such relations, alongside STRING [3] and OmniPath [4]. Because curation is slow and expensive, there is strong interest in recovering relations automatically, first with biomedical language models [5], [6] and relation-extraction corpora [7], and now with general-purpose large language models (LLMs) [8]–[13].

In this line of work KEGG serves as the answer key: a model receives a gene pair, names the relation type, and is scored against the curated label. High scores are then read as evidence that the model *knows* pathway biology or that a prompting strategy helps it reason. That reading is valid only if the benchmark does not reward shortcuts. For pair-input prediction it often does: when test pairs share members with training pairs, performance is inflated by node-level regularities rather than pair-level understanding [14]–[16]. Related concerns have been raised for gene regulatory network validation [17], and more generally as data leakage [18] and shortcut learning [19].

KEGG relation benchmarks add two complications that have not been examined. First, labels are typed and co-occur: a kinase can both phosphorylate and activate its substrate. Second, genes are named by HGNC symbols [20], whose family stems (*MAPK*, *PRKA*, *DUSP*) carry functional information that any model, including an LLM pre-trained on biomedical text, can exploit without knowing anything about the specific pair.

Here we ask what a KEGG gene–gene relation benchmark actually measures. Fig. 1 summarizes the study. Our contributions are:

- **A structural audit** of a public benchmark released with a recent LLM prompt-optimization study [13], quantifying label co-annotation, cross-pathway redundancy and pair overlap, and the ceiling these impose on single-label accuracy.
- **A shortcut demonstration**: a model that reads only gene symbols matches the performance band reported for LLMs under the random-row protocol.
- **A leakage-aware reformulation** as pair-level multi-label prediction with four splits of increasing strictness, and an evaluation of seven predictors, from a non-learned lookup to a character-level convolutional network (CharCNN) and an inductive graph neural network (GNN).
- **A mechanistic decomposition** showing that pair-disjoint scores are explained by seen genes, that relation mechanism transfers across pathways through kinase-family stems while relation sign does not, and that this gap is not caused by inconsistent sign annotation.
- **A biomedical language-model arm**: four open-source pre-trained encoders (BioBERT v1.1 and v1.0, SciBERT and SPECTER), used as frozen feature extractors and, for BioBERT, fine-tuned end to end, evaluated under the same splits and with a scrambled-name control.
- **Four stress tests**, namely symbol scrambling, direction

swapping, calibration under distribution shift and an overlap analysis of held-out pathways, together with model ensembling, precision–recall analysis, split-conformal uncertainty quantification, curation ranking and a label-permutation control, which confirm that the models rely on name spelling and gene roles and become overconfident exactly where pathway knowledge is required.

All numbers are produced by released code from public data and can be regenerated on a laptop CPU.

II. Related Work

A. AI for Pathway Knowledge

Domain-specific encoders such as BioBERT [5] and PubMedBERT [6] established strong baselines for relation extraction from text, and corpora such as BioRED [7] define typed relations at document level. With LLMs, attention has shifted to whether models can produce pathway relations from parametric knowledge. Evaluations have covered molecular interaction and pathway extraction [9], [10], chatbot-assisted biological pathway curation [11], broad biomedical NLP tasks [8], knowledge extraction from pathway figures [12], and automated prompt search with tree-structured Bayesian optimization [13], [21], and multi-modal LLM–graph models of kinase–substrate–phosphatase regulation [22]. All rely on a trusted reference, typically KEGG.

B. Evaluation Pitfalls for Pair-Input Prediction

Park and Marcotte [14] showed that pair-input predictors tested on pairs whose members appear in training are dominated by component-level effects, and proposed stratifying test pairs by how many members were seen. Hamp and Rost [15] and Bernett *et al.* [16] documented similar inflation in protein interaction prediction, where deep models exploited node degree and sequence similarity. Tabe-Bordbar *et al.* [17] showed that random cross-validation overestimates gene regulatory network accuracy, Kapoor and Narayanan [18] catalogued leakage across machine-learning-based science, and Geirhos *et al.* [19] framed shortcut learning in general.

C. Link Typing in Networks

Predicting the type or sign of an edge from network structure has a long history [23], [24] and has been applied to biomedical graphs, including typed drug–drug edges [25] and protein interactions via path-based similarity [26]. Node embeddings [27] and inductive message passing [28] are standard inputs to such predictors. We use them here as probes of what information in the benchmark suffices to score well.

III. Data and Structural Audit

A. Source

We used the KEGG gene-to-gene relation tables introduced in [13]: one table per KEGG human pathway listing a starter (source) symbol, a receiver (target) symbol and a relation name. The 23 tables cover MAPK, ErbB, Ras, Rap1, cGMP–PKG, cAMP, mTOR, AMPK, p53, apoptosis (two maps), necroptosis, autophagy, endocytosis, longevity, oocyte meiosis, cytokine and chemokine signaling, endoplasmic-reticulum protein processing, adrenergic signaling in cardiomyocytes, and three drug-resistance pathways (EGFR tyrosine kinase inhibitor, endocrine and platinum). Although stored with a `.csv` extension, the files are spreadsheets; we read all 23 and concatenated them with a pathway tag.

TABLE I
Structural Audit of the KEGG-Derived Benchmark

Quantity	Value
Records (all types) / retained	3,207 / 3,166
Pathway tables / distinct genes	23 / 860
Exact duplicate records within a pathway	192
Duplicate (starter, receiver, relation) triples	557
Unique directed gene pairs	2,176
Pairs with > 1 label	426 (19.6%)
Pairs in > 1 pathway (max.)	244 (10)
Gene families (suffix rule)	432
Random 70/20/10 row split, 20 seeds	
Test rows whose pair is in training	$41.0 \pm 1.6\%$
Test rows whose exact triple is in training	$22.9 \pm 1.1\%$
Upper bound on single-label accuracy	0.842

B. Composition

The combined data contain 3,207 records (Table I, Fig. 2a). Activation (1,426), inhibition (903) and phosphorylation (837) account for 98.7%; the remaining 41 records of five rare types were excluded, leaving 3,166 records over 860 genes. Label mixes differ sharply between pathways: the two cytokine maps contain only activation, AMPK and oocyte meiosis are dominated by phosphorylation, and apoptosis by inhibition.

C. Label Co-Annotation, Redundancy and Overlap

Collapsing records to unique directed pairs shows that 426 of 2,176 pairs (19.6%) carry more than one label (Fig. 2b), nearly always phosphorylation together with a sign. This is biologically expected, because phosphorylation describes the *mechanism* of an interaction and activation or inhibition its *effect*, and KEGG annotates both. Treating records as independent single-label instances therefore makes some answers ambiguous by construction; a predictor forced to output one label per pair can reach at most 0.842 accuracy. Cross-pathway redundancy and within-pathway duplication (Table I) mean that a random row split places the same pair, and often the same (pair, label) triple, on both sides of the split.

IV. Methods

A. Task Formulations

Task R (row-level, single-label) reproduces current practice: each record is an instance with one of three labels, split at random 70/20/10 into training, test and an unused validation set, as in [13]. We also report a pair-disjoint variant in which all records of an unordered pair fall into one partition. Each setting is repeated with 10 seeds.

Task M (pair-level, multi-label) is our proposed formulation: for each unique directed pair (s, r), predict $\mathbf{y} \in \{0, 1\}^3$



Fig. 1. Study overview. (a) Twenty-three curated KEGG human signaling maps provide 3,207 typed, directed relations among 860 genes (schematic; arrow, activation; bar, inhibition; dashed arrow, phosphorylation). (b) The audit reveals four properties hidden by random row splits: co-annotated pairs, test rows whose pair was already seen in training, informative gene-name families, and a ceiling on single-label accuracy. (c) Four leakage-aware splits of increasing strictness (red blocks). (d) Seven machine-learning predictors read either the two gene symbols or the training graph, and four pre-trained biomedical language models (BioBERT v1.1 and v1.0, SciBERT, SPECTER) encode the symbols; each outputs three label probabilities per pair (bars are illustrative). (e) Stress tests, measured values across pair (P), gene (G), family (F) and pathway (Pw) splits: LR-Lex AUROC with original versus scrambled symbols and original versus swapped gene roles, expected calibration error of LR-Lex and GBDT-Graph, and coverage (%) of 90% split-conformal prediction sets. (f) Per-label AUROC of the symbol-only LR-Lex model: phosphorylation transfers to unseen pathways, whereas activation and inhibition fall toward chance.

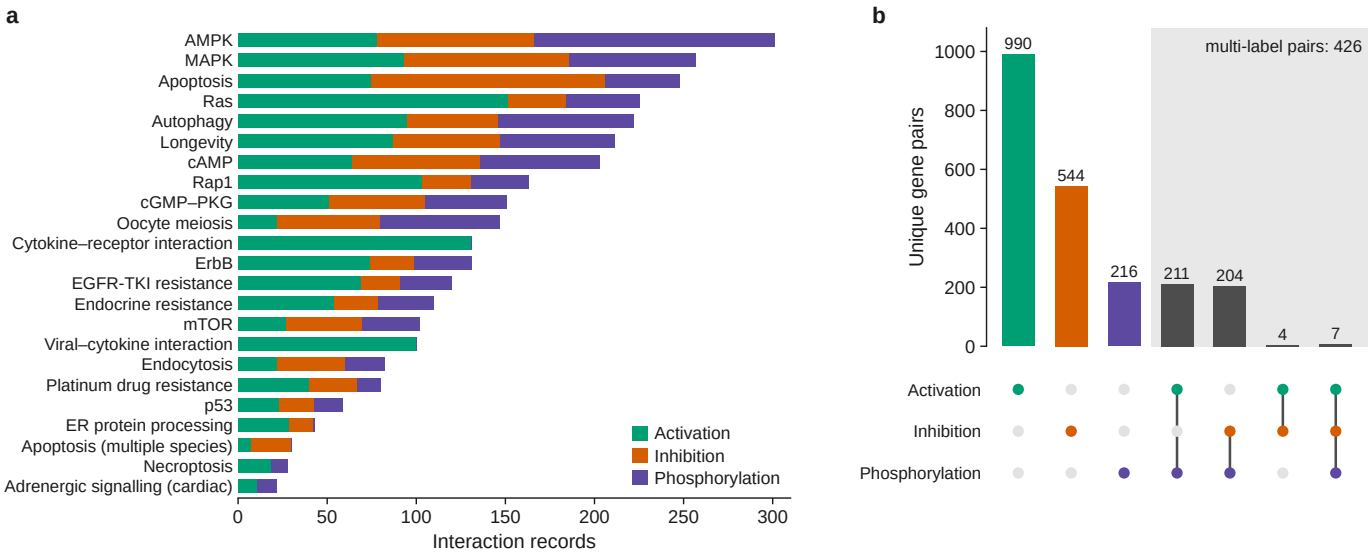


Fig. 2. Benchmark composition. (a) Relation records per KEGG pathway, colored by type; composition differs strongly between pathways. (b) Label sets of the 2,176 unique directed gene pairs. Colored bars are single-label pairs; dark bars to the right of the dotted line are the 426 multi-label pairs, almost all of which combine a sign with phosphorylation.

indicating activation, inhibition and phosphorylation [29]. This removes duplicates, represents mechanism and sign explicitly, and makes splits unambiguous.

B. Splitting Regimes

Task M is evaluated under four regimes (Fig. 1c). All features and model parameters are learned from training pairs

only.

1) *Pair-disjoint*: unordered pairs are assigned to five folds (two repetitions), so a pair and its reverse never straddle the split.
2) *Gene-disjoint*: genes are assigned to five folds (two repetitions); test pairs involve at least one held-out gene and training pairs none, corresponding to the C2/C3 classes of [14].
3) *Family-disjoint*: as above, but folds are formed over gene families obtained by stripping the trailing numeric suffix of each symbol (e.g., *MAPK1*, *MAPK3* $\rightarrow$ *MAPK*), giving 432 families (largest: CCL, 27 genes). This removes paralog shortcuts.
4) *Pathway-held-out*: each of the 23 pathways is held out in turn; the test set is every pair annotated in it, and training excludes all such pairs in either orientation.

C. Predictors

Prior predicts training label prevalence. *Lookup* is a non-learned node-profile heuristic: the score for label ℓ averages the smoothed frequency $\hat{p}_\ell(g) = (n_{g\ell} + \pi_\ell)/(n_g + 1)$ with which s carries ℓ as a starter and r as a receiver in training, where π_ℓ is the label prevalence.

LR-Lex encodes the two symbols as TF–IDF-weighted character 2–4-grams in separate vocabularies and fits one class-balanced logistic regression per label ($C = 1$). It sees no text, prompt or pathway context.

LR-Graph and *GBDT-Graph* use 48 features of the directed training graph: smoothed starter and receiver label profiles with support counts; in-degree, out-degree and PageRank [30] of both genes; the number of shared intermediates; and a rank-16 truncated SVD embedding of the adjacency matrix with the starter–receiver inner product. They are fitted by logistic regression or by histogram gradient boosting [31] (300 iterations, learning rate 0.05, 15 leaves). *LR-Lex+Graph* concatenates both feature sets.

CharCNN is a neural symbol model in the spirit of [32]: each symbol is embedded character by character (24 dimensions), passed through parallel 1-D convolutions of width 2, 3 and 4 (64 filters each) with max pooling, separately for starter and receiver, and the two codes are combined by a two-layer perceptron with dropout 0.3.

GNN is an inductive two-layer GraphSAGE network [28] with mean aggregation (64 hidden units) over the undirected training graph. Node inputs are a 64-dimensional SVD projection of each symbol's character n-grams plus log degree, so genes absent from training still receive representations. A pair is scored by a perceptron on $[\mathbf{h}_s, \mathbf{h}_r, \mathbf{h}_s \odot \mathbf{h}_r, |\mathbf{h}_s - \mathbf{h}_r|]$.

Both neural models are trained full-batch with AdamW [33] (learning rate 3×10^{-3} and 5×10^{-3}, weight decay 10^{-4}, 250 and 300 epochs) on a class-weighted binary cross-entropy, with fixed hyperparameters across all regimes so that no test information is used for tuning.

For Task R we use a multinomial LR-Lex ($C = 3$), a majority-class baseline, a starter-lookup that returns the starter's most frequent training label (backing off to the receiver, then the global majority), and a pair-lookup that first returns the most frequent training label of the same pair.

D. Evaluation

For Task R we report macro-F1 and the Matthews correlation coefficient (MCC) [34]. For Task M we report macro-averaged AUROC, which is threshold-free, and macro-F1 (threshold 0.5 for learned models and training prevalence for Prior and Lookup), plus per-label AUROC. Prior predicts every label as positive and scores 0.54–0.56 macro-F1, so macro-F1 should be read against it; precision–recall analysis [35] is possible from the released predictions. Pathway-held-out results are test-size-weighted means over pathways, with 95% confidence intervals from 5,000 bootstrap resamples of pathways; the two single-label cytokine pathways have undefined AUROC and are excluded from AUROC averages. Paired comparisons across folds use the two-sided Wilcoxon signed-rank test [36].

E. Stress Tests

Four additional analyses probe what the models rely on, using the same folds. *Symbol scrambling* replaces every gene symbol by a random six-character token (one fixed token per gene, so identity is preserved but spelling is destroyed) and retrains LR-Lex. *Direction swap* scores each test pair with starter and receiver exchanged and compares the predictions with the original labels; a model that uses gene roles should lose accuracy. *Calibration* is measured by the expected calibration error (ECE, 10 equal-width bins, averaged over labels) [37] and the Brier score. *Overlap* is the fraction of gene occurrences in a held-out pathway that also occur in the training pairs, related to per-pathway AUROC by Spearman correlation.

F. Ensembling, Uncertainty and Ranking

Five further analyses use the same folds. *Ensemble* averages the probabilities of LR-Lex, LR-Graph and GBDT-Graph [38]. *AUPRC* is the macro-averaged area under the precision–recall curve [35], compared with label prevalence. *Split-conformal prediction* [39], [40] holds out a random 20% of the training pairs for calibration, refits LR-Lex on the rest, and forms per-label prediction sets with nonconformity $1-\hat{p}(y)$ at a nominal 90% coverage; we report empirical coverage on the test fold, which is guaranteed only if test pairs are exchangeable with calibration pairs. *Curation ranking* measures precision among the ten highest-scoring test pairs per label (precision@10), divided by the label's prevalence in that fold. A *label-permutation control* [41] retrains LR-Lex after shuffling the label vectors of the training pairs; any score above 0.5 would indicate leakage in the pipeline itself. Experiments used scikit-learn [42], NetworkX [43], SciPy [44] and PyTorch [45] on a CPU.

G. Biomedical Language Models

To test whether knowledge acquired from the literature helps where symbol spelling does not, we added four open-source BERT-style encoders [46]: BioBERT v1.1, pre-trained on PubMed abstracts, and BioBERT v1.0, pre-trained on PubMed abstracts and PubMed Central full texts [5]; SciBERT, pre-trained on scientific papers [47]; and SPECTER, a SciBERT

TABLE II
TASK R: ROW-LEVEL SINGLE-LABEL RESULTS (MEAN ± S.D., 10 SEEDS)

Split	Model	Macro-F1	MCC
Random row	Majority	0.208 ± 0.008	0.000
	Pair-lookup	0.628 ± 0.014	0.466 ± 0.022
	Starter-lookup	0.691 ± 0.016	0.558 ± 0.021
	LR-Lex	**0.731** ± 0.011	**0.614** ± 0.016
Pair-disjoint	Majority	0.208 ± 0.004	0.000
	Starter-lookup	0.699 ± 0.026	0.575 ± 0.042
	LR-Lex	**0.780** ± 0.009	**0.688** ± 0.014

model further trained on citation links between papers [48]. Weights were taken from the authors' public releases. Each model has 12 layers and 110 million parameters.

In the *frozen* setting each gene symbol is encoded on its own, and its vector is the mean over its word pieces of the last four hidden layers (768 dimensions). A pair is represented by $[\mathbf{e}_s, \mathbf{e}_r, \mathbf{e}_s \odot \mathbf{e}_r]$, standardized and scored by one class-balanced logistic regression per label ($C = 0.05$). In the *fine-tuned* setting BioBERT v1.1 reads the pair as a sentence pair, "[CLS] s [SEP] r [SEP]", and is trained end to end with a three-output sigmoid head on the [CLS] token, using AdamW (learning rate 3×10^{-5}, weight decay 0.01), batch size 32, four epochs with linear decay and class-weighted binary cross-entropy. Fine-tuning used the five folds of the first seed for the pair-, gene- and family-disjoint regimes and all pathway folds; frozen models used every fold. A *scrambled-name control* embeds each gene's fixed random six-letter token (Section IV-E) instead of its symbol, so that the encoder cannot recognize the gene. No setting was tuned on test data.

V. RESULTS

A. Gene Symbols Alone Match Reported LLM Performance

Under the random-row protocol, LR-Lex reaches a macro-F1 of 0.731 and an MCC of 0.614 from gene symbols alone (Table II). The study that released this benchmark reported macro-F1 between about 0.55 and 0.80 for five LLMs under the same row-level protocol [13]. We did not rerun those models and the samples differ, so the numbers are not directly comparable, but the overlap shows that the protocol cannot by itself separate biological knowledge from symbol-level regularities.

LR-Lex scores *higher* under the pair-disjoint split (0.780) than under the random split. A per-row diagnosis explains why: across 10 seeds, 58.8% of random-split test rows have an unseen pair (accuracy 0.886), 22.7% have a training copy with the same label (0.819), and 18.6% have a training copy with a *different* label (0.209). In the last group the model reproduces the training copy and is marked wrong because the pair is co-annotated. Random-row splitting therefore inflates scores through overlap and deflates them through label conflict at the same time, and its results cannot be interpreted cleanly in either direction.

TABLE III
TASK M: MACRO AUROC / MACRO-F1 UNDER FOUR SPLITS. MEAN OVER FOLDS (PATHWAY: TEST-SIZE-WEIGHTED MEAN OVER 23 PATHWAYS). BEST AUROC PER COLUMN IN BOLD.

Model	Pair	Gene	Family	Pathway
Prior	.500/.562	.500/.560	.500/.561	.500/.540
Lookup	.974/.863	.891/.738	.785/.652	.597/.484
LR-Lex	.978/.909	**.939**/.844	**.811**/.688	.683/.509
LR-Graph	.982/.922	.871/.686	.763/.555	.620/.447
LR-Lex+Graph	.983/.926	.878/.691	.770/.559	.609/.445
GBDT-Graph	**.987**/.931	.878/.712	.776/.600	.606/.471
CharCNN	.986/.949	.929/.838	.771/.651	**.697**/.520
GNN	**.987**/.948	.895/.772	.779/.631	.674/.510
Test pairs / fold	435	784	769	13–248

B. Performance Collapses as Splits Tighten

Table III summarizes Task M and Fig. 3a shows the spread across folds and pathways. When only pairs are held out, every learned model exceeds 0.97 macro AUROC, the neural models reach 0.986–0.987, and even the non-learned Lookup reaches 0.974. When one partner is unseen (gene-disjoint), the graph models fall to 0.87–0.90, whereas the two symbol models hold at 0.93–0.94 because unseen genes usually have seen paralogs. Holding out whole families removes this advantage: all models fall to 0.76–0.81 and the spread between them narrows to within fold variability. Under pathway hold-out every model approaches the floor; the best is CharCNN at 0.697 (95% CI 0.659–0.735), followed by LR-Lex at 0.683 (0.640–0.724) and the GNN at 0.674 (0.622–0.729), and no model exceeds the Prior on macro-F1.

The more expressive AI models do not escape the pattern. CharCNN improves on LR-Lex by 0.008 AUROC when pairs are held out ($P = 0.027$) but is 0.010 and 0.040 worse under gene- and family-disjoint splits (both $P = 0.002$). LR-Lex outperforms GBDT-Graph in every regime except pair-disjoint (gene $+0.062$, family $+0.035$, pathway $+0.068$; all $P \leq 0.002$), while GBDT-Graph is slightly better when pairs are held out (-0.008, $P = 0.002$). Model capacity thus helps exactly where shortcuts are available and does not help where they are removed. Per-pathway results for LR-Lex (Fig. 4) range from 0.44 (adrenergic signaling in cardiomyocytes, 13 pairs) to 0.82 (EGFR-TKI resistance), with a median of 0.68, and exceed chance in 20 of 21 evaluable pathways.

C. Seen Genes, Not Seen Pairs, Drive Pair-Disjoint Scores

Stratifying pair-disjoint test pairs by how many of their genes appear in training [14] (Fig. 5a) shows that most test pairs (on average 372 of 435 per fold) have both genes seen; on these, Lookup already reaches 0.977 AUROC and GBDT-Graph 0.990. Pairs with one seen gene (57 per fold) score 0.96 for all three models, and only about 12 pairs per fold have no seen gene, too few to evaluate. A pair-disjoint score therefore mostly measures how well a model recalls what its two genes did elsewhere. The gene-disjoint learning curve (Fig. 5b) makes the same point from another angle: with only 10% of the training pairs (about 140), LR-Lex already reaches 0.851 AUROC, whereas the graph-based Lookup and LR-Graph need all the data to reach 0.87–0.89.

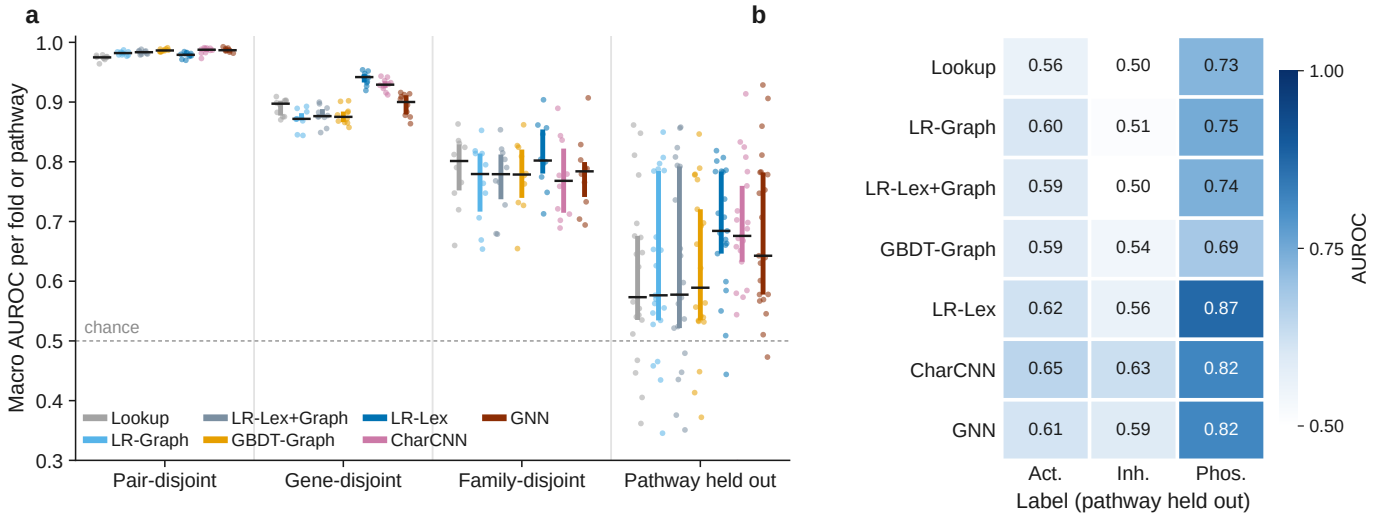


Fig. 3. Variability and per-label behavior under four splits (means are given in Table III). (a) Macro AUROC of the seven predictors on every fold (pair-, gene- and family-disjoint; 10 folds each) and every evaluable held-out pathway (21); bars span the interquartile range and black ticks mark the median. The spread grows as the split tightens, and under pathway hold-out every model scores below chance on at least one pathway (minimum 0.30–0.47) and at most 0.82–0.93 on the best one. (b) Per-label AUROC of every predictor when whole pathways are held out (test-size-weighted means): phosphorylation transfers, whereas activation and inhibition stay near chance for all models.

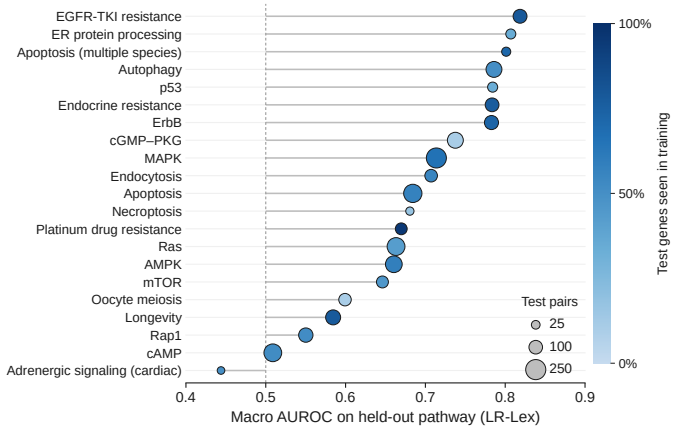


Fig. 4. Macro AUROC of LR-Lex on each held-out pathway (21 evaluable pathways; the two single-label cytokine pathways are omitted). Marker area is proportional to the number of test pairs and marker shade to the fraction of test-gene occurrences seen in training; the dashed line marks chance.

D. Mechanism Transfers; Sign Does Not

The collapse under pathway hold-out is not uniform across labels (Fig. 3b and Fig. 1f). Phosphorylation remains predictable in unseen pathways (AUROC 0.82–0.87 for the symbol and neural models), whereas activation and inhibition fall to 0.56–0.65 for every model. The learned weights explain why (Fig. 5d). Phosphorylation is driven by starter-gene n-grams that mark kinase families, such as the prefixes of *MAPK*/*MAP2K* and *PRK*-family kinases; a model can recognize a kinase by name in any pathway. Inhibition draws on prefixes of deubiquitinases (*USP*) and cathepsins (*CTS*) and on adenylyl-cyclase-like receivers, and activation on growth factors and calmodulin; these are weaker, more pathway-specific cues.

E. Sign Failure Is Not Annotation Noise

A natural objection is that sign is simply inconsistent across pathways, so no model could predict it. The data do not support this. Among the 244 pairs annotated in more than one pathway, the sign is identical in every pathway for 83% and inhibition status for 95% (Fig. 5c); only 7 of 219 pairs with a sign in several pathways carry opposite signs. The sign of a known pair is therefore largely stable, yet the sign of a *new* pair in an *unseen* pathway cannot be inferred from names or from the topology of other pathways. This is precisely the knowledge a benchmark of pathway understanding should measure, and it is the part that current benchmark designs leave untested.

F. Stress Tests Confirm Reliance on Names and Roles

The stress tests (Fig. 6) show what the models use. Scrambling symbol spelling barely affects pair-disjoint AUROC (-0.009), where gene identity suffices, but costs 0.089 under gene-disjoint, 0.065 under family-disjoint and 0.094 under pathway hold-out (all $P < 0.01$, Wilcoxon). The lexical model's advantage over graph models on unseen genes (Table III) therefore comes from the spelling of gene names. Swapping starter and receiver lowers pair-disjoint AUROC from 0.978 to 0.655, so the models encode gene roles, such as kinases acting as starters, rather than symmetric similarity.

Calibration degrades with the strictness of the split. GBDT-Graph is the best calibrated model on pair-disjoint data (ECE 0.039 against 0.111 for LR-Lex), yet the worst on held-out pathways (0.373 against 0.235; $P < 10^{-5}$). Its Brier score rises eightfold, from 0.042 to 0.362, whereas that of LR-Lex rises from 0.065 to 0.222. Topology-based models are therefore confident precisely where they are wrong, which matters if their probabilities are used to rank candidate interactions for curation.

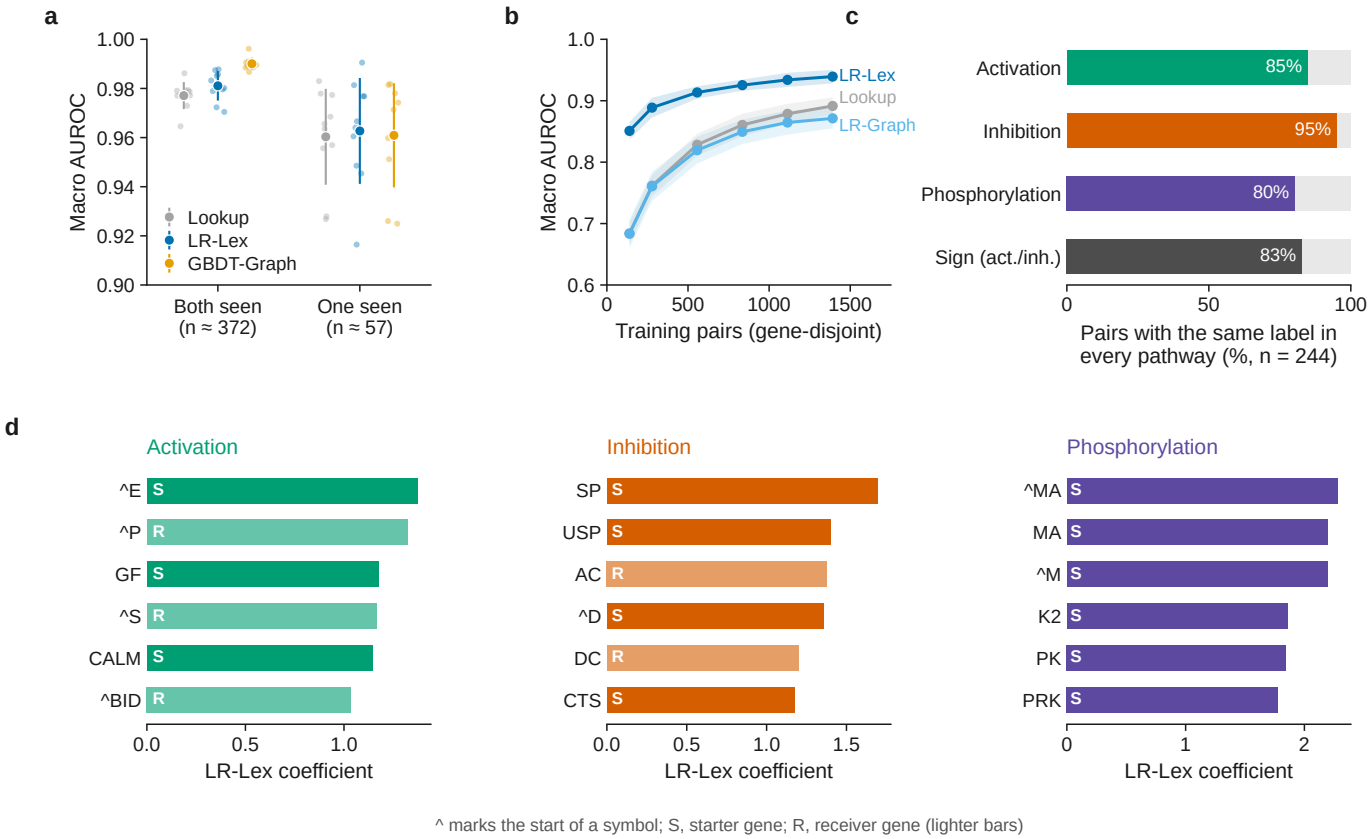


Fig. 5. Where the signal comes from. (a) Pair-disjoint AUROC stratified by the number of genes of each test pair seen in training; small points are folds. (b) Gene-disjoint learning curves; shading, one standard deviation over 10 folds. (c) Fraction of the 244 pairs annotated in more than one pathway whose label is identical in every pathway. (d) The six largest LR-Lex coefficients per label, fitted on all pairs; prefixes of kinase families dominate phosphorylation.

Finally, difficulty of a held-out pathway is not explained by how many of its genes were seen. The fraction of test-gene occurrences seen in training is 92% for pair-disjoint, 41% for gene-disjoint, 36% for family-disjoint and 55% for pathway-held-out folds, and across held-out pathways it ranges from 11% to 96%, yet its correlation with LR-Lex AUROC is weak and non-significant (Spearman $\rho = 0.15$, $P = 0.52$; GBDT-Graph $\rho = 0.37$, $P = 0.10$; Fig. 4). Seeing a gene in other pathways does not reveal how it acts in a new one, which is consistent with the sign result above.

G. Ensembles, Uncertainty and Ranking Do Not Rescue Transfer

Fig. 7 extends the analysis to ensembling, uncertainty quantification and practical use. Averaging three models gives the best pair-disjoint AUROC (0.988, $P = 0.002$ against LR-Lex), but no gain where pathway knowledge is needed: it is below LR-Lex on gene-disjoint (0.924 against 0.939, $P = 0.002$) and pathway-held-out data (0.659 against 0.683, $P = 0.05$), and indistinguishable on family-disjoint data ($P = 0.32$). Model diversity therefore amplifies the shortcut rather than replacing it.

Precision–recall analysis confirms the ranking from AUROC. Macro AUPRC of LR-Lex falls from 0.962 to 0.645 on held-out pathways, against a prevalence baseline of 0.43, GBDT-Graph falls from 0.978 to 0.596 and the ensemble from 0.980 to 0.630 (Fig. 7a).

Split-conformal prediction, which promises a coverage of at least 90% under exchangeability, meets this target only on pair-disjoint data (90.6%). Coverage drops to 85.1% for unseen genes, 72.8% for unseen families and 58.0% for unseen pathways (all $P \leq 0.002$ against 90%, Wilcoxon), with individual pathways as low as 26% (Fig. 7b). Uncertainty estimates calibrated on a random subset of the benchmark therefore cannot be trusted for new pathways.

For curation, the ten top-ranked pairs per label are 1.7–3.4 times enriched over prevalence on pair-, gene- and family-disjoint data, but only 1.16 times for activation and 1.30 times for inhibition on held-out pathways, while phosphorylation keeps a 2.13-fold enrichment (Fig. 7c). A curator using these models to prioritize activating or inhibiting interactions in a new pathway would gain little over random order.

The label-permutation control scores 0.49–0.52 AUROC under every split (Fig. 7d), so the signal reported above is genuine, and the losses come from the absence of transferable knowledge rather than from errors in the evaluation pipeline.

H. Pre-trained Biomedical Language Models Inherit the Shortcut

Table IV and Fig. 8 report the language-model arm. Frozen representations from all four encoders reach the same AU-

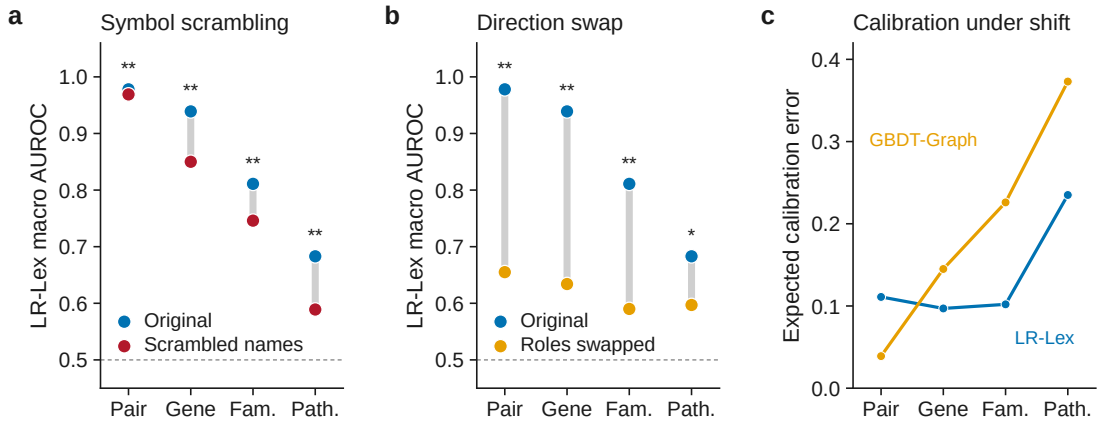


Fig. 6. Stress tests. (a) LR-Lex macro AUROC with original and scrambled gene symbols (identity kept, spelling destroyed). (b) LR-Lex AUROC when starter and receiver are swapped at test time. Asterisks in (a) and (b) give two-sided Wilcoxon signed-rank tests over folds: $*P < 0.05$, $**P < 0.01$. (c) Expected calibration error across splits. Dashed lines mark chance.

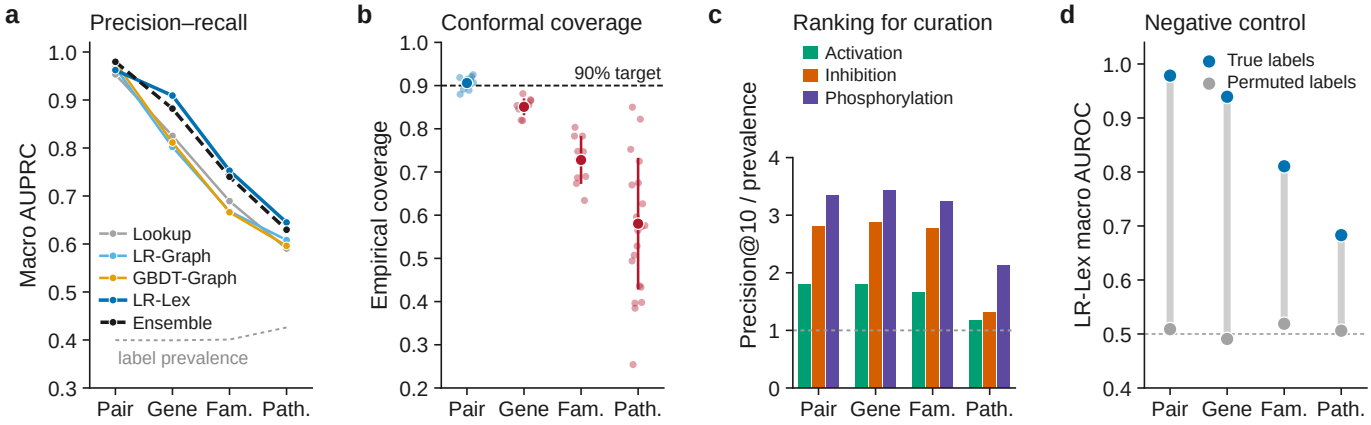


Fig. 7. Ensembling, uncertainty and ranking. (a) Macro AUPRC across splits for four models and their ensemble; the dashed grey line is label prevalence. (b) Empirical coverage of split-conformal prediction sets at a nominal 90% level (dashed line); small points are folds or held-out pathways, large points are means with one standard deviation. (c) Precision among the ten top-ranked test pairs per label, divided by prevalence; the dashed line marks random ranking. (d) LR-Lex AUROC with true and with permuted training labels; the dashed line marks chance.

TABLE IV
PRE-TRAINED BIOMEDICAL LANGUAGE MODELS UNDER THE FOUR SPLITS: MACRO AUROC AND PER-LABEL AUROC ON HELD-OUT PATHWAYS. PATHWAY VALUES ARE TEST-SIZE-WEIGHTED MEANS; BOLD MARKS THE BEST LANGUAGE MODEL PER COLUMN. THE SYMBOL-ONLY REFERENCE (LR-LEX) IS IN TABLE III AND FIG. 3B.

	Macro AUROC by held-out level				Pathway held out, per label		
Model	Pair	Gene	Family	Pathway	Act.	Inh.	Phos.
BioBERT v1.1, frozen	0.985	0.935	0.798	0.707	0.626	**0.659**	0.837
BioBERT v1.0, frozen	**0.986**	**0.937**	0.798	0.698	0.636	0.633	0.826
SciBERT, frozen	**0.986**	0.935	0.816	0.706	0.629	0.658	0.834
SPECTER, frozen	0.979	0.927	0.808	0.697	0.616	0.613	**0.866**
BioBERT v1.1, fine-tuned	0.981	**0.937**	**0.831**	**0.732**	**0.675**	0.656	**0.866**

ROC as the symbol-only baseline at every level of the split hierarchy. On pair-disjoint data they are marginally higher (0.979–0.986 against 0.978); on unseen genes and families they are within 0.015 of LR-Lex; and on held-out pathways they reach 0.697–0.707 against 0.683, with no model significantly better ($P \geq 0.41$, Wilcoxon over pathways; Fig. 8a). Fine-tuning BioBERT end to end gives 0.981, 0.937, 0.831 and 0.732 (95% CI 0.67–0.79) across the four levels. It is indistinguishable from LR-Lex on pair-, gene- and family-disjoint data ($P \geq 0.06$, five folds) and is the only model in this study that is significantly better on held-out pathways (+0.049, $P = 0.046$, Wilcoxon over pathways; better on 14 of 21 pathways, Fig. 8a), although its confidence interval still overlaps that of LR-Lex. Pre-training on the literature therefore adds at most a small amount of knowledge that transfers to new pathways.

The per-label pattern is the one seen for every other model (Table IV). On held-out pathways the language models reach

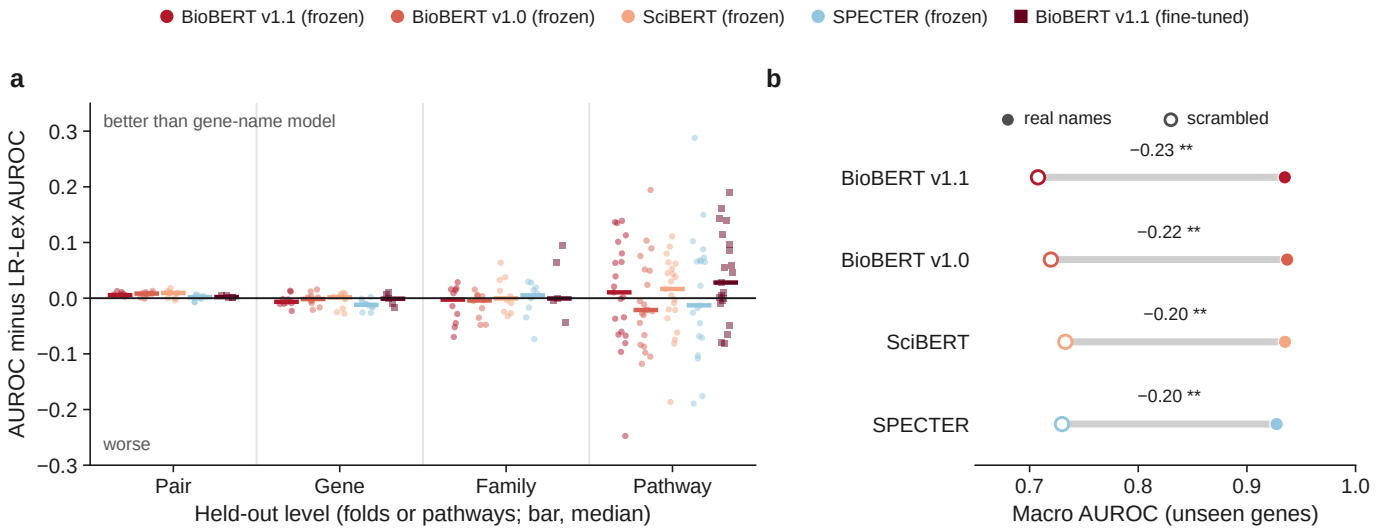


Fig. 8. Pre-trained biomedical language models inherit the gene-name shortcut (means are given in Table IV). (a) Difference in macro AUROC between each language model and the symbol-only LR-Lex model on the same fold or held-out pathway; bars mark medians. Differences stay near zero at every level and spread widely across pathways. (b) On unseen genes, replacing symbols by random tokens (open circles) lowers the AUROC of every frozen model by 0.20–0.23 (Wilcoxon, ** $P < 0.01$).

0.83–0.87 for phosphorylation but only 0.62–0.68 for activation and 0.61–0.66 for inhibition, so the regulatory sign remains far from the 0.98 reached when pairs are only held out. Fine-tuning raises activation and inhibition to 0.68 and 0.66 (LR-Lex 0.62 and 0.56), which accounts for its small overall gain, but leaves the sign well below phosphorylation (0.87).

The scrambled-name control shows where the language-model signal comes from. On unseen genes, embedding random tokens instead of the real symbols lowers AUROC from 0.93–0.94 to 0.71–0.73 (all $P = 0.002$; Fig. 8b). This loss is more than twice the 0.09 lost by LR-Lex under the same control, so the encoders depend even more on recognising gene names. What they recognize is the identity and family of a gene, which helps only where training data already describe a similar gene.

VI. Discussion

A. Implications for AI and LLM Knowledge Mining

Our results do not show that generative LLMs lack pathway knowledge, because we evaluated encoder language models rather than generative LLMs. They show that the benchmark, as commonly used, cannot establish that such models have it, and that four biomedical language models pre-trained on the literature behave much like a classifier of gene-name spelling: used as frozen encoders they add nothing, and fine-tuned they add a few points of AUROC that leave the regulatory sign largely unresolved. Under random-row splitting a symbol-only classifier scores in the range attributed to LLM knowledge or prompt design, and the score is simultaneously inflated by pair overlap and deflated by co-annotation. An LLM is not trained on the split, but it was pre-trained on text that names the same gene families, so it has access to the same lexical regularities. Two observations make this concrete: within our own experiments, adding capacity (CharCNN, GNN) helped only where shortcuts existed, and the label that transfers across pathways is exactly the one whose cue is in the gene name. Prompt-optimization studies face an additional risk, because prompts selected on a validation set drawn from the same random split can overfit to pair- and family-level regularities [18].

B. Recommendations

We recommend that studies using KEGG-derived relation benchmarks to evaluate AI systems:

1) deduplicate to unique directed pairs and treat the task as multi-label, or report sign and mechanism as separate sub-tasks;
2) report family-disjoint and pathway-held-out results alongside any random split, together with the fraction of test pairs whose genes or families were seen;
3) include a symbol-only baseline (e.g., LR-Lex), a scrambled-symbol control applied to the AI system itself and a node-profile baseline (Lookup) as floors that an AI system must beat;
4) report calibration (ECE or Brier score) and, where uncertainty is claimed, conformal coverage under each split, because confidence on unseen pathways is unreliable;
5) report per-label metrics, because pooled scores are dominated by the comparatively easy phosphorylation label;
6) select prompts or hyperparameters on validation data that are disjoint from the test set at the same level as the claim being made.

All splits in this study are generated deterministically from seeds in the released code so that future LLM evaluations can use identical partitions.

C. Limitations

We analyzed one public benchmark of 23 human pathways; other KEGG extractions, species or relation vocabularies may

differ, although the mechanisms identified here are properties of KEGG annotation and HGNC naming rather than of this extraction. Our family definition is a suffix rule that groups most paralogs but misses irregularly named families; HGNC family assignments [20] would be more precise. We excluded 1.3% of records with rare relation types. Neural hyperparameters were fixed rather than tuned, which avoids test leakage but may understate their best attainable performance. Our language-model arm used four open 110-million-parameter encoders, frozen or fine-tuned on CPU; larger generative LLMs, prompted or fine-tuned, may behave differently, and we cannot quantify how much of a particular LLM's reported score comes from the shortcuts described here. Evaluating generative LLMs under the proposed splits is the most important next step.

VII. Conclusion

High scores on KEGG-derived relation benchmarks can be obtained from gene names alone. Once label co-annotation, pair overlap and gene-family naming are controlled, every model we tested, including a character-level CNN, a graph neural network, their ensemble and four pre-trained biomedical language models, falls toward chance when predicting whether a new interaction activates or inhibits its target in an unseen pathway; only phosphorylation, which kinase names reveal, transfers. Scrambling gene names costs the language models about 0.2 AUROC, and stress tests confirm that all models rely on name spelling and gene roles and become overconfident where they fail, with split-conformal coverage dropping from 91% to 58% on held-out pathways. AI and LLM knowledge mining should therefore be evaluated per label under family- and pathway-disjoint splits, with calibration and symbol-only baselines. Future work will use the released splits to evaluate generative LLMs, test whether biological evidence beyond gene names, such as sequence, structure or literature, restores transfer of regulatory sign, and extend the benchmark to other pathway resources and species.

Data and Code Availability

The combined relation table, split definitions, analysis code, figure scripts and per-fold results generated in this study are available at https://github.com/Muh-aza/aaa.

References

[1] M. Kanehisa and S. Goto, "KEGG: Kyoto Encyclopedia of Genes and Genomes," *Nucleic Acids Res.*, vol. 28, no. 1, pp. 27–30, 2000, doi: 10.1093/nar/28.1.27.

[2] M. Kanehisa, M. Furumichi, Y. Sato, M. Kawashima, and M. Ishiguro-Watanabe, "KEGG for taxonomy-based analysis of pathways and genomes," *Nucleic Acids Res.*, vol. 51, no. D1, pp. D587–D592, 2023, doi: 10.1093/nar/gkac963.

[3] D. Szklarczyk, R. Kirsch, M. Koutrouli, K. Nastou, F. Mehryary, R. Hachilif *et al.*, "The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest," *Nucleic Acids Res.*, vol. 51, no. D1, pp. D638–D646, 2023, doi: 10.1093/nar/gkac1000.

[4] D. Türei, T. Korcsmáros, and J. Saez-Rodriguez, "OmniPath: guidelines and gateway for literature-curated signaling pathway resources," *Nat. Methods*, vol. 13, no. 12, pp. 966–967, 2016, doi: 10.1038/nmeth.4077.

[5] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020, doi: 10.1093/bioinformatics/btz682.

[6] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, "Domain-specific language model pretraining for biomedical natural language processing," *ACM Trans. Comput. Healthcare*, vol. 3, no. 1, pp. 2:1–2:23, 2022, doi: 10.1145/3458754.

[7] L. Luo, P.-T. Lai, C.-H. Wei, C. N. Arighi, and Z. Lu, "BioRED: a rich biomedical relation extraction dataset," *Brief. Bioinform.*, vol. 23, no. 5, p. bbac282, 2022, doi: 10.1093/bib/bbac282.

[8] Q. Chen, Y. Hu, X. Peng, Q. Xie, Q. Jin, A. Gilson *et al.*, "Benchmarking large language models for biomedical natural language processing applications and recommendations," *Nat. Commun.*, vol. 16, no. 1, p. 3280, 2025, doi: 10.1038/s41467-025-56989-2.

[9] G. Park, B.-J. Yoon, X. Luo, V. López-Marrero, S. Yoo, and S. Jha, "Comparative performance evaluation of large language models for extracting molecular interactions and pathway knowledge," arXiv:2307.08813, 2023, doi: 10.48550/arXiv.2307.08813.

[10] M. Azam, Y. Chen, M. O. Arowolo, H. Liu, M. Popescu, and D. Xu, "A comprehensive evaluation of large language models in mining gene relations and pathway knowledge," *Quant. Biol.*, vol. 12, no. 4, pp. 360–374, 2024, doi: 10.1002/qub2.57.

[11] M. Azam, F. He, KRPNS Santosh, D. Xu, and M. Popescu, "Evaluation and integration of advanced AI chatbots for biological pathway curation," in *Proc. IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, Lisbon, Portugal, 2024, pp. 6342–6348, doi: 10.1109/BIBM62325.2024.10822181.

[12] M. Azam, S. Zeng, H. Aldihis, M. Popescu, and D. Xu, "Applications of large language models and prompt optimization for knowledge extraction from biological pathway figures," *IEEE J. Biomed. Health Inform.*, vol. 30, no. 2, pp. 1680–1692, 2026, doi: 10.1109/JBHI.2025.3631538.

[13] M. Azam, H. Aldihis, S. Zeng, and D. Xu, "Bayesian optimization of ASCII structural anchors for improving large language model performance in biomedical knowledge mining," Research Square preprint, 2026, doi: 10.21203/rs.3.rs-9243236/v1.

[14] Y. Park and E. M. Marcotte, "Flaws in evaluation schemes for pair-input computational predictions," *Nat. Methods*, vol. 9, no. 12, pp. 1134–1136, 2012, doi: 10.1038/nmeth.2259.

[15] T. Hamp and B. Rost, "More challenges for machine-learning protein interactions," *Bioinformatics*, vol. 31, no. 10, pp. 1521–1525, 2015, doi: 10.1093/bioinformatics/btu857.

[16] J. Bernett, D. B. Blumenthal, and M. List, "Cracking the black box of deep sequence-based protein–protein interaction prediction," *Brief. Bioinform.*, vol. 25, no. 2, p. bbae076, 2024, doi: 10.1093/bib/bbae076.

[17] S. Tabe-Bordbar, A. Emad, S. D. Zhao, and S. Sinha, "A closer look at cross-validation for assessing the accuracy of gene regulatory networks and models," *Sci. Rep.*, vol. 8, no. 1, p. 6620, 2018, doi: 10.1038/s41598-018-24937-4.

[18] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023, doi: 10.1016/j.patter.2023.100804.

[19] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nat. Mach. Intell.*, vol. 2, no. 11, pp. 665–673, 2020, doi: 10.1038/s42256-020-00257-z.

[20] R. L. Seal, B. Braschi, K. Gray, T. E. M. Jones, S. Tweedie, L. Haim-Vilmovsky, and E. A. Bruford, "Genenames.org: the HGNC resources in 2023," *Nucleic Acids Res.*, vol. 51, no. D1, pp. D1003–D1009, 2023, doi: 10.1093/nar/gkac888.

[21] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2019, pp. 2623–2631, doi: 10.1145/3292500.3330701.

[22] M. O. Arowolo, M. E. Okon, D. Austria, M. Azam, and S. O. Abdulsalam, "SPINET-KSP: A multi-modal LLM-graph foundation model for contextual prediction of kinase-substrate-phosphatase triads," *Kinases and Phosphatases*, vol. 4, no. 1, p. 3, 2026, doi: 10.3390/kinasesphosphatases4010003.

[23] D. Liben-Nowell and J. Kleinberg, "The link-prediction problem for social networks," *J. Am. Soc. Inf. Sci. Technol.*, vol. 58, no. 7, pp. 1019–1031, 2007, doi: 10.1002/asi.20591.

[24] J. Leskovec, D. Huttenlocher, and J. Kleinberg, "Predicting positive and negative links in online social networks," in *Proc. 19th Int. Conf. World Wide Web*, 2010, pp. 641–650, doi: 10.1145/1772690.1772756.

[25] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, vol. 34, no. 13, pp. i457–i466, 2018, doi: 10.1093/bioinformatics/bty294.

[26] I. A. Kovács, K. Luck, K. Spirohn, Y. Wang, C. Pollis, S. Schlabach *et al.*, "Network-based prediction of protein interactions," *Nat. Commun.*, vol. 10, no. 1, p. 1240, 2019, doi: 10.1038/s41467-019-09177-y.

[27] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2016, pp. 855–864, doi: 10.1145/2939672.2939754.

[28] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 1024–1034.

[29] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, 2014, doi: 10.1109/TKDE.2013.39.

[30] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Comput. Netw. ISDN Syst.*, vol. 30, no. 1–7, pp. 107–117, 1998, doi: 10.1016/S0169-7552(98)00110-X.

[31] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[32] X. Zhang, J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification," in *Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 28, 2015, pp. 649–657.

[33] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[34] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 1, p. 6, 2020, doi: 10.1186/s12864-019-6413-7.

[35] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015, doi: 10.1371/journal.pone.0118432.

[36] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945, doi: 10.2307/3001968.

[37] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 1321–1330, https://proceedings.mlr.press/v70/guo17a.html.

[38] T. G. Dietterich, "Ensemble methods in machine learning," in *Multiple Classifier Systems (MCS 2000)*, ser. Lecture Notes in Computer Science, vol. 1857. Springer, 2000, pp. 1–15, doi: 10.1007/3-540-45014-9_1.

[39] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, 2nd ed. Cham, Switzerland: Springer, 2022, doi: 10.1007/978-3-031-06649-8.

[40] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023, doi: 10.1561/2200000101.

[41] M. Ojala and G. C. Garriga, "Permutation tests for studying classifier performance," *Journal of Machine Learning Research*, vol. 11, pp. 1833–1863, 2010, https://www.jmlr.org/papers/v11/ojala10a.html.

[42] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[43] A. A. Hagberg, D. A. Schult, and P. J. Swart, "Exploring network structure, dynamics, and function using NetworkX," in *Proc. 7th Python Sci. Conf. (SciPy 2008)*, 2008, pp. 11–15.

[44] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nat. Methods*, vol. 17, no. 3, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2.

[45] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 8024–8035.

[46] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT)*, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[47] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in *Proc. Conf. Empirical Methods Natural Lang. Process. and Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, 2019, pp. 3613–3618, doi: 10.18653/v1/D19-1371.

[48] A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. Weld, "SPECTER: Document-level representation learning using citation-informed transformers," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2020, pp. 2270–2282, doi: 10.18653/v1/2020.acl-main.207.